

HealSphere: An Open Source-Based Mental Health Support Platform



This real-world CI/CD implementation has been developed using open source tools to deploy a modular mental health support platform.

Designing a reliable, scalable platform for mental health support came with its fair share of engineering challenges. From the beginning, we weren't just focused on features or frontend polish; we knew the real work would be in building the DevOps pipeline that could support multiple services, continuous updates, and real-time monitoring without crumbling under load.

This article is a behind-the-scenes look at how we built and deployed HealSphere, not from a product perspective, but from an infrastructure point of view. Using a mix of Spring Boot microservices, a React frontend, and a toolchain that includes Docker, Kubernetes, Jenkins, Ansible, and the ELK Stack, we assembled a CI/CD pipeline that supports automated builds, container orchestration, and centralised logging.

We'll walk through the real decisions we made, the rough edges we hit, and how we tied everything together using open source tools.

Architecture overview and key layers

From the start, we designed HealSphere as a modular, full-stack microservices application. The platform supports three user roles: patients, doctors, and administrators, each interacting with a distinct set of services. On the backend, functionality is split across several Spring Boot based microservices, while the frontend is developed using React. All components are containerised, orchestrated with Kubernetes, and tied together using a fully automated CI/CD pipeline.

In the high-level architecture of HealSphere, each service is independently containerised using Docker and deployed to a Kubernetes cluster via Ansible-based automation scripts, triggered through a Jenkins CI/CD pipeline. We maintained a separate container for each microservice, which simplifies scaling and debugging, especially when traffic spikes unevenly across features (e.g., appointment scheduling vs resource access).